HENRI WITHÖFT

Physics & Aerospace Engineering Aspirant · Abitur Scholar

Düsseldorf, Germany
Email: Henri.withoeft@gmail.com
GitHub: github.com/henri20HUB
LinkedIn: linkedin.com/in/henri-withöft-2a106b338

ACADEMIC PROFILE & OBJECTIVES

Abitur candidate with an international academic background focused on Physics, Aerospace/Electrical Engineering, and Computational Problem-Solving. Targeting undergraduate studies at **ETH Zürich**, **EPFL**, and **TU München**.

EDUCATIONAL TIMELINE

Comenius-Gymnasium Düsseldorf **Aug. 2026 – May 2028 (Expected)**
Abitur Candidate (Upper Secondary Education) *Düsseldorf, Germany*
Focusing on advanced mathematics, natural sciences, and analytical writing tracks.

Pocklington School **Aug. 2025 – June 2026**
GCSE Qualifications & Academic Exchange *Yorkshire, UK*
Curriculum emphasizing analytical thinking, physics, mathematics, and computer science in an English-speaking environment.

Heinrich-Heine-Universität Düsseldorf **Sept. 2024 – March 2025**
Schülerstudium (Early University Studies) *Düsseldorf, Germany*
Attended university lectures and conducted coursework in Medical Physics and Experimental Mechanics.

Comenius-Gymnasium Düsseldorf **2018 – July 2025**
ZP10 Abschluss (Secondary Education Qualification) *Düsseldorf, Germany*

ADVANCED CERTIFICATIONS & COURSEWORK

Space Mission Design and Operations **Issued Oct. 2026**
EPFL (École Polytechnique Fédérale de Lausanne) • edX Verified *Credential ID: 69310ea121934419a1554bffe41eec6a*
Covered orbital mechanics, launch dynamics, propulsion systems, spacecraft subsystems, and mission trade-off analysis.

CS50x: Introduction to Computer Science **Issued May 2026**
Harvard University *Credential ID: 433365b7-e595-4dfa-8bb4-b5b364f66201*
Mastered fundamental computer science concepts, memory management (C), data structures, algorithms, SQL, and web development.

TECHNICAL PROJECTS & RESEARCH

Multi-Axis Robotic Arm Build **In Progress**
Developing a physical robotic arm utilizing Raspberry Pi microcomputers and Arduino microcontrollers for motor actuation and sensor control.

Automated Portfolio Infrastructure **2026**
Engineered a multi-page web presence utilizing pure HTML, Tailwind CSS, Git version control, and Vercel edge deployment.

PUBLICATIONS & HUMANITIES RESEARCH

Co-Author — "Die Unglaubliche Geschichte von Rosa Willinger" **Published June 18, 2025**
Edition Virgines • ISBN / Historical Research Project
Co-authored a published historical book examining local Jewish history in the Rhineland, participating in the Geschichtswettbewerb des Bundespräsidenten (Körber-Stiftung) with primary archival evaluation.

SKILLS & LANGUAGES

Languages:	German (Native), English (Fluent), French (Basic)
Technical / STEM:	Orbital Mechanics, Computational Mechanics, Python, C, SQL, HTML/Tailwind, Microcontrollers (Raspberry Pi, Arduino)
Research:	Source-Critical Analysis, Archival Investigation, Scientific Writing